6.1.3 Charging and discharging capacitors

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	**(i)** charging and discharging capacitor through a resistor	
	(ii) techniques and procedures to investigate the charge and the discharge of a capacitor using both meters and data-loggers	**PAG9** HSW4 Investigating the charge and discharge of capacitors in the laboratory.
(b)	time constant of a capacitor–resistor circuit; $\tau = CR$	HSW9
(c)	equations of the form $x = x_0 e^{-\frac{t}{CR}}$ and $x = x_0(1 - e^{-\frac{t}{CR}})$ for capacitor–resistor circuits	Learners will be expected to know how ln$x$–t graphs can be used to determine *CR*. *M0.5, M2.5, M3.10, M3.12*
(d)	graphical methods and spreadsheet modelling of the equation $\frac{\Delta Q}{\Delta t} = -\frac{Q}{CR}$ for a discharging capacitor	HSW3 Using spreadsheets to model the discharge of a capacitor. *M3.9*
(e)	exponential decay graph; constant-ratio property of such a graph.	*M3.11*